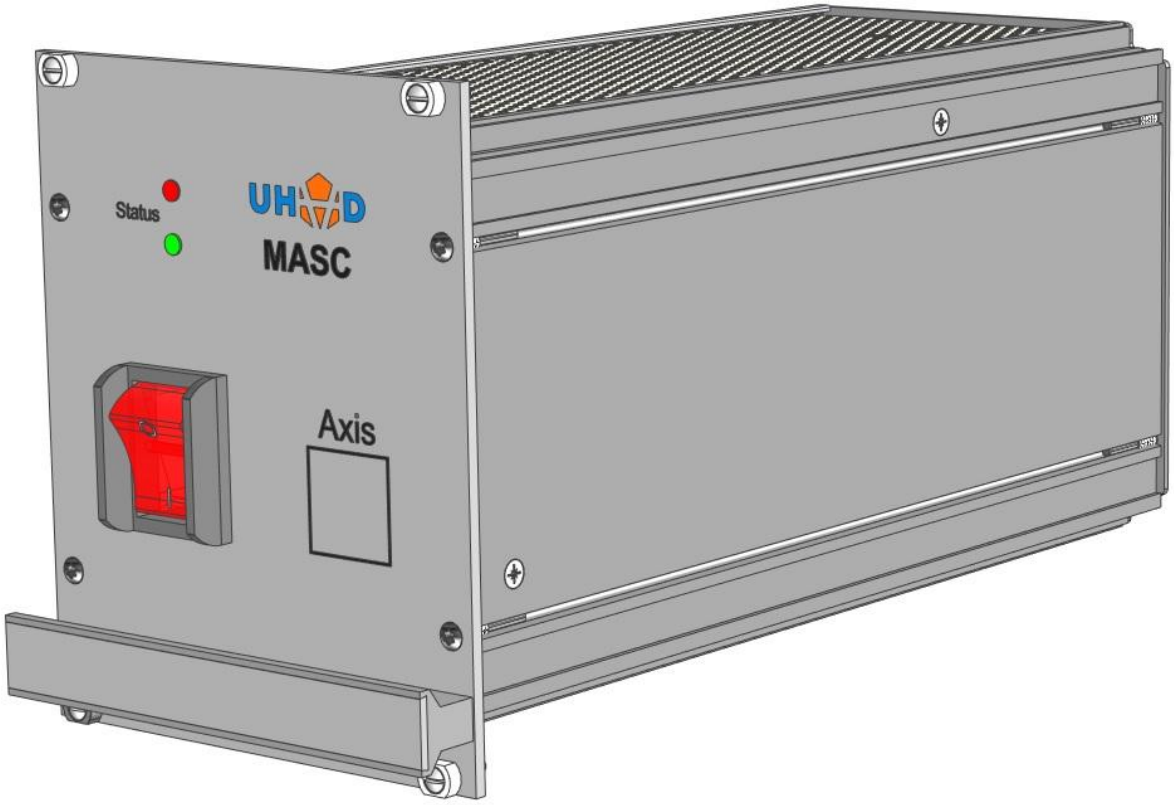
PRODUCT INSTRUCTION MANUAL

Product: MASC Stepper Motor Controller



Product Description

Stepper Motor Controller

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1 IMPORTANT SAFETY INFORMATION!

This document is intended as a user guide to operation as determined by the design.
Position control systems are inherently hazardous. Even a small motor, if coupled to a leadscrew, gearbox, or any other form of mechanism that provides a mechanical advantage, can generate considerable force and could cause serious injury. Incorrect operation can also lead to damage to the motor or associated machinery. It is essential that the purpose of the various fault detection features built into the MASC's ST5-Q controller be fully understood and used correctly.

As such, Installation must be carried out only by competent persons who are familiar with electronic control systems.

Product Standard	BS EN IEC 61800-5-1:2023
Protective Class	C1
Ingress Protection	IP20
EMC Class	C1*
Maximum conditional short circuit current	5kA

***When using a MASC-JB jog box in a domestic environment, this product may cause radio interference, in which case supplementary mitigation measures may be required.**

	WARNING: Risk of electric shock — Capacitor discharge time 30 s. Allow 30 s before unplugging any cables or attempting maintenance.
	This product can cause a DC current in the PE conductor Where a residual current-operated protective device (RCD) is used for protection against electric shock, only an RCD of Type B is allowed on the supply side of this product. All upstream RCD, up to the supply transformer, shall be of Type B

Associated Documents:

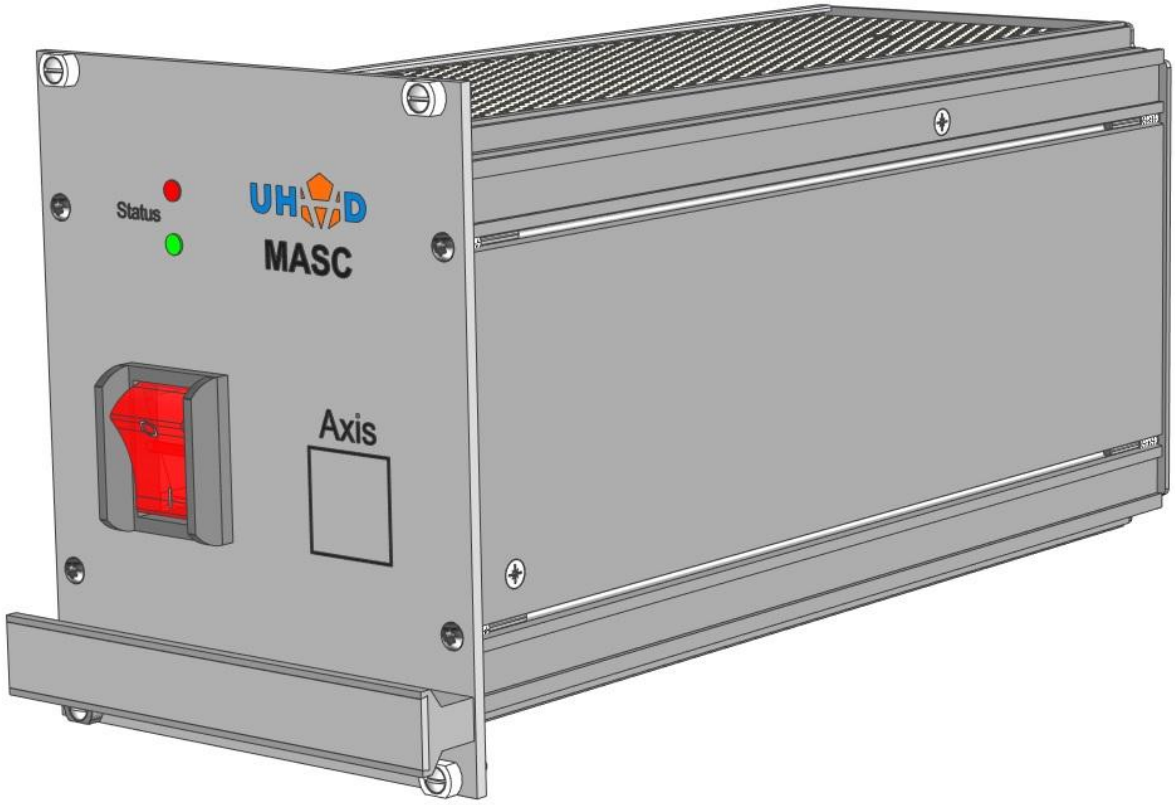
ST5-Q Manual
Host Command Reference Manual

We reserve the right to update the data used in this manual in line with product development without prior notice.

2 DESCRIPTION

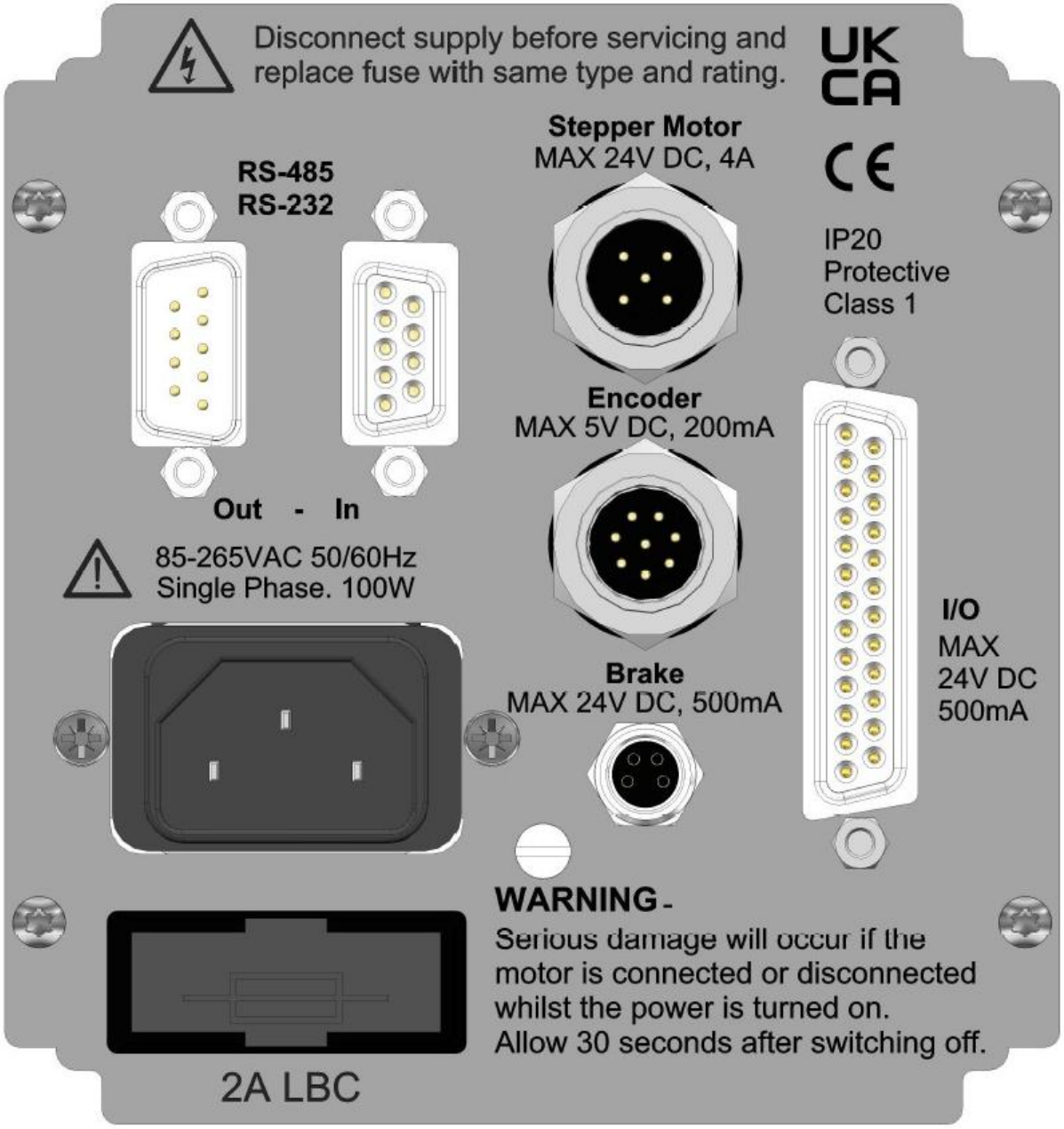
The ***MASC*** is a single axis stepper motor controller in a compact rack mountable enclosure. It is a versatile, high-performance microstepping packaged unit suitable for light industrial and scientific uses where precision and system-level integration are required. Monitoring and control can be achieved over a RS-485 serial connection that can be easily configured in multi-drop for the coordinated use of several drivers over the same connection. RS232 is also an option for single axis applications.

The MASC controller consists of a ST5-Q programmable microstepping drive and an integrated 24V DC universal input power supply built into a 21HP 3U subrack unit for mounting into industry standard 19" racks. This allows for up to four MASC units to be fitted in one standard 84HP wide rack. The MASC is supplied with a mains lead, an RS485 cable and an RS485 terminator. All connections to and from the unit are on the rear panel and an illuminated on-off switch with two status indicators are easily accessible on the front panel. A range of standard motor, encoder, limits/datum and I/O cables are available.



An incremental encoder can be fitted to the motor or mechanism to monitor the movement of the mechanism. The quadrature signals from the encoder are connected to the ST5-Q to allow position monitoring or stall prevention (if the encoder is mounted on the motor shaft).
The power supply provides +24V (nominal) to energise the supply on the ST5-Q controller, and a low current +24V supply for the I/O connector. The I/O supply is protected by a 0.5A non replaceable self-resetting fuse.

There are two diagnostic STATUS LED's shown on the front panel. These indicate the status of the controller and any error conditions that may be present, see section 5 for alarm code table or refer to the Host Command Reference documentation for a description of the different patterns.



2.1 Mains input.

Mains input is via a filtered IEC inlet.

The MASC uses a Switched Mode Power Supply to provide the Stepper Motor Driver with DC power. As such it is designed to accept a wide range of single phase AC voltages, from 85 to 265V, with no hardware changes, at frequencies ranging from 47Hz to 63Hz. Each driver is supplied with a mains lead with an IEC C13 connector to mate with an IEC C14 back-panel power supply connector. (The mains lead has a 5A fuse and the MASC has a replaceable 2A mains fuse on the rear panel. UK, EU and US cables are supplied with the unit).

General Specifications	
Max input power	100W

WARNING! To avoid overheating, the vents in the top and bottom of the unit must be unobstructed at all times.

2.2 Motor output.

The Stepper Motor is connected to the MASC driver using an M12 A-coded, 5-pin Female connector. Maximum motor output current is limited to 4A. Max output voltage is 24V. No motor overtemperature protection is provided.

General motor specifications:	
Max phase current	4A
Max voltage	24V

WARNING! Serious damage will occur if the motor is connected or disconnected whilst the power is switched on. Allow 30 seconds after switching off.

2.3 Encoder input.

The encoder connector is an M12 A-coded, 8-pin Female connector.

It has a +5V supply to energise the encoder, complementary quadrature inputs and complementary index inputs.

A wide range of 5V rotary and linear encoders can be supported.

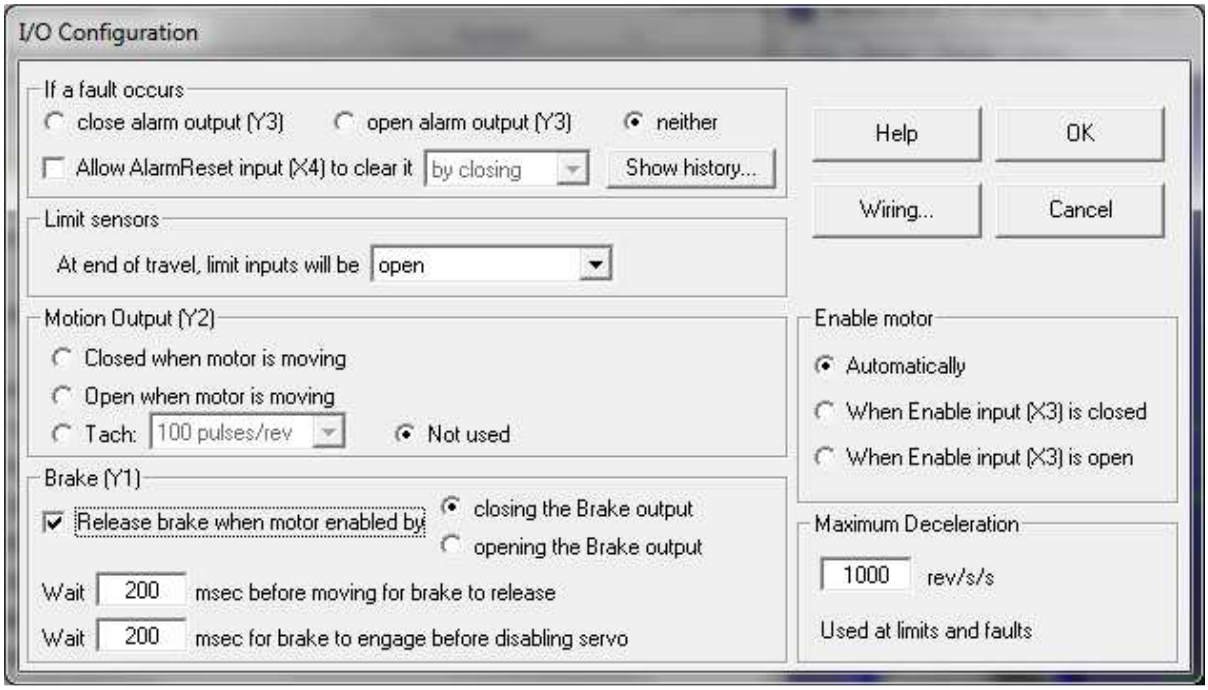
The use of single-ended encoders is not encouraged.

The Encoder connector on the MASC Drivers are directly pin compatible with the Nanotec IP65 family of Stepper Motors that have built in Encoders (https://en.nanotec.com/products/154-ip65-stepper-motors/).

Note: if external encoders are installed by the customer, then insulation level must be maintained so as not to cause damage

2.4 Brake Output

The Brake connector is an M8 A-coded, 4-pin Female connector.
It provides a 24V DC, 0.5A fused supply for electrically operated mechanical brakes.
It is controlled by the Y1/Brake signal on pin 14 of the I/O connector, and hence may be automatically controlled with enabling/disabling and fault conditions of the drive.
The drive should be configured for brake operation using the ST-Configurator program, under the I/O box. It should be enabled and set for "Release brake when motor enabled by – closing the Brake output". The timings may be configured here too.



Alternatively use the **BO** (BO1), **BD** and **BE** commands.
See the host command reference manual.

2.5 I/O connector.

The I/O connector is a standard 25 pin female D connector.
Summary of the I/O functions available (note that some may be mutually exclusive):

- CW & CCW Limit Switching in both current sourcing and sinking configurations
- Home position sensing
- Programmable Digital Outputs
- Analogue and Digital inputs
- Alarm signalling
- Direct (non-serial) Jog Control

A full description of the I/O capabilities of the ST5-Q driver unit can be found in the Applied Motion Products Hardware and Host Command Reference documents available from the Applied Motion Products website. The links to these documents are provided in Section 5.
NOTE: The I/O connector differs in one important respect from the Applied Motion Products ST5-Q

25-pin D-sub pinout. Pin 3 which is a spare pin on the ST5-Q product, has 24V applied to it with a 500mA non replaceable self-resetting fuse that is internal to the MASC (shown in Red in the IO connections of section 2). This modification is made available so that limit switches can be used with the MASC, removing the need for other external 24V power arrangements at the limit switches. It is up to the user to ensure that the additional 24V DC output is used appropriately and within the current limit imposed for the protection of cabling and the driver unit.

Any limit switches should be normally closed, opening on detection of a limit condition, this will also stop the motor if any of the cables or wiring become damaged or disconnected.

Limit switches must not be used as a safety device or part of a safety system for ensuring the safety of persons.

The following two figures illustrate the connectivity for the use of Normally Closed Limit Switches using the 24V supply in Pin 3 to allow current to flow through the switch and the optically isolated Driver Input circuit. This configuration allows for PNP limit sensor in an Open Collector configuration as in Figure 3, resulting in a current sourcing connection to the I/O line.

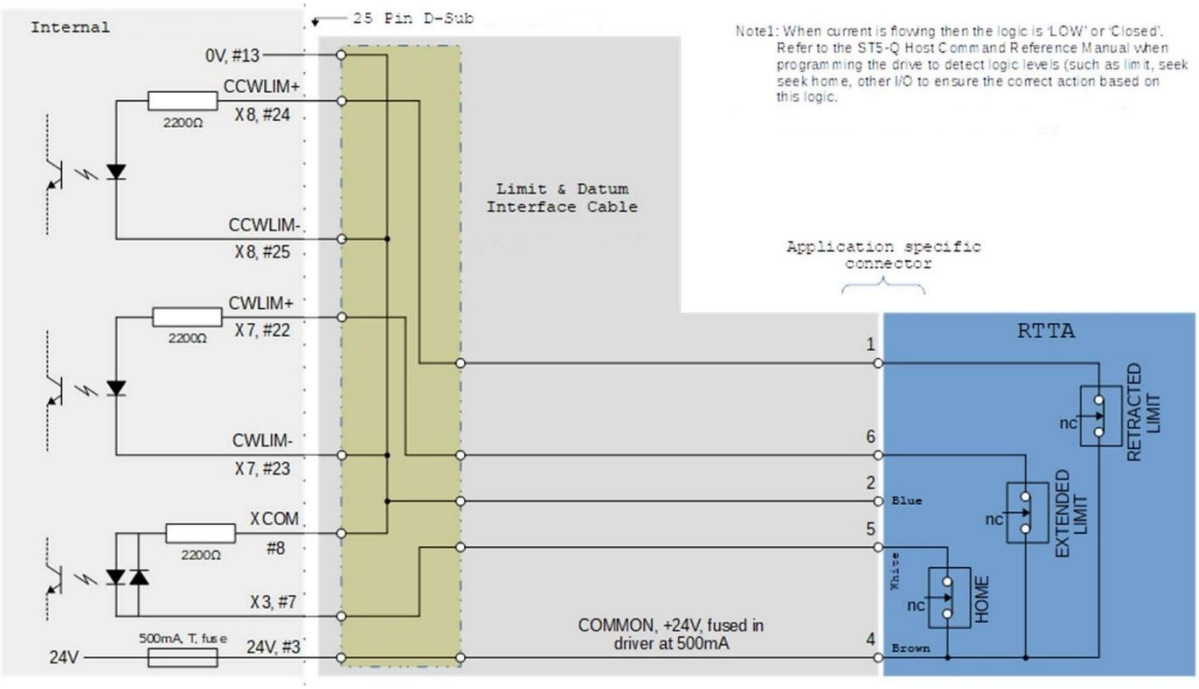


Figure 2 – Limits detection for mechanical (NC) Limit Switches

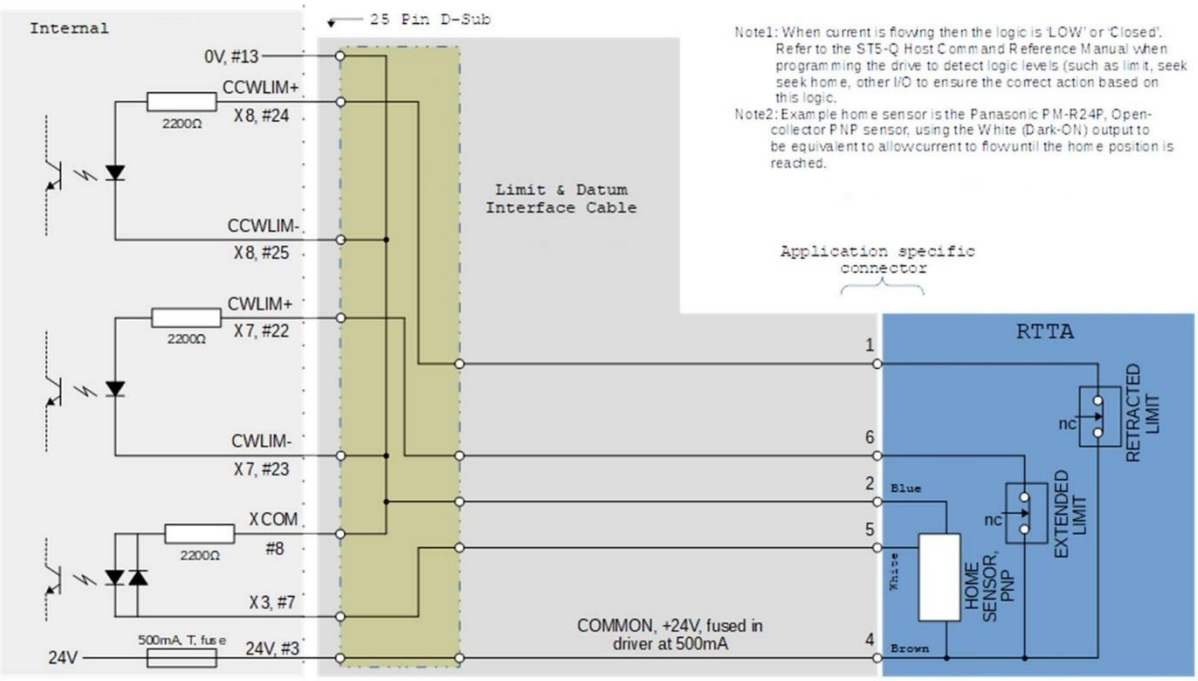


Figure 3 –Limits for mechanical (NC) Limit Switches and a PNP Home Switch

2.6 RS-485 Multi drop

RS-485 Multidrop connectors are provided to allow a dedicated serial connection to a small network of MASC drivers. This can allow the control and monitoring of multiple drivers from a single RS-485 Master such as a Handheld Jog Unit, PC or logic controller device
In order for RS-485 to function reliably a termination resistor is required at the end of the Multidrop network furthest from the Master. A termination resistor module MASC-TP can be attached to the final driver's male RS-485 OUT port.
Example of RS485 connection from a PC to a single MASC using an RS485-USB converter.

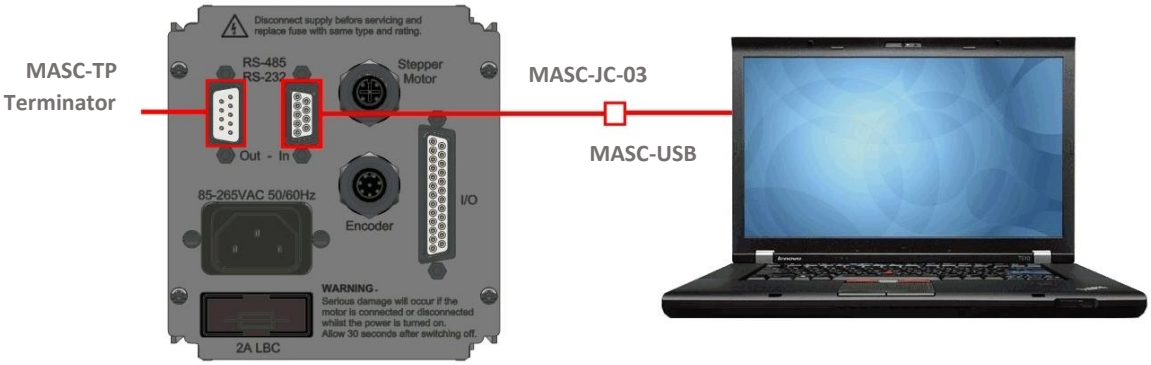


Figure 4 – RS485 Connection with USB Converter

Example of an RS485 connection to multiple ***MASC*** units (using smart jog box MASC-JB).

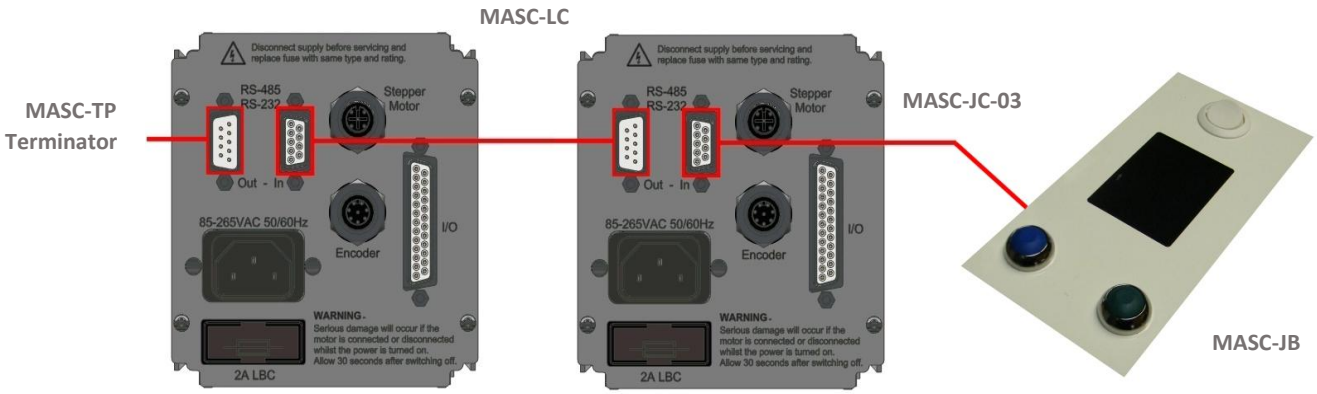


Figure 5 – RS485 Multidrop with Intelligent Jog box

2.7 RS232 For single axis operation or configuration.

Example of RS232 connection from a PC.

Figure 6 – RS232 connection from a PC

3 External Interface Connections

MAINS	
Connector type: IEC chassis plug	
Terminal	**Connection**
L	Live input 85 - 265V AC (Universal input) Max input power 100W
N	Neutral input
E	Earth. **THIS UNIT MUST BE EARTHED**

MOTOR	
Connector type: M12 A-coded, 5-pin Female connector.	
Max motor output current limited to 4A.	
Max output voltage limited to 24V	
Mating Cable: MASC-MC-05 or MASC-MC-10	
Terminal	**Connection**
1	Motor Phase A-
2	Motor Phase A+
3	Motor Phase B+
4	Motor Phase B-
5	Screen

BRAKE	
Connector type: M8 A-coded, 4-pin Female connector.	
Max output current 0.5A	
Mating Cable: MASC-BC-xx01	
Terminal	**Connection**
1	Brake 0V
2	Brake +24V DC out **(500mA max)**
3 & 4	No connection

I/O	
Connector type: 25 way 'D' Socket	
Mating Cable: MASC-LDC-xxxx	
Terminal	**Connection**
1	Analog input 1
2	Analog input 2
3	**+24V DC out (500mA max)**
4	X6 / CCw Jog
5	X5 / Cw Jog
6	X4 / Alarm reset
7	X3 / Enable
8	X common
9	X2- / DIR-
10	X2+ / DIR+
11	X1- / STEP-
12	X1+ / STEP+
13	Gnd
14	Y1 / Brake *
15	Y2 / Motion
16	Y3 / Fault
17	Y common **
18	+5V out ***
19	Gnd
20	Y4+
21	Y4-
22	X7+ / Cw Limit+
23	X7- / Cw Limit-
24	X8+ / CCw Limit+
25	X8- / CCw Limit-

*Implicitly connected to the Brake output amplifier. If isolation is required then contact Manufacturer
** Implicitly connected to the supply 0V. If isolation is required then contact Manufacturer
***Maximum capacity 200mA shared between this connector and the I/O connector.

ENCODER	
Connector type: M12 A-coded, 8-pin Female connector	
Mating Cable: MASC-ENC-xxxx	
Terminal	**Connection**
1	Channel A+
2	Channel A-
3	Channel B+
4	Channel B-
5	0V
6	Channel I-
7	Channel I+
8	+5V **

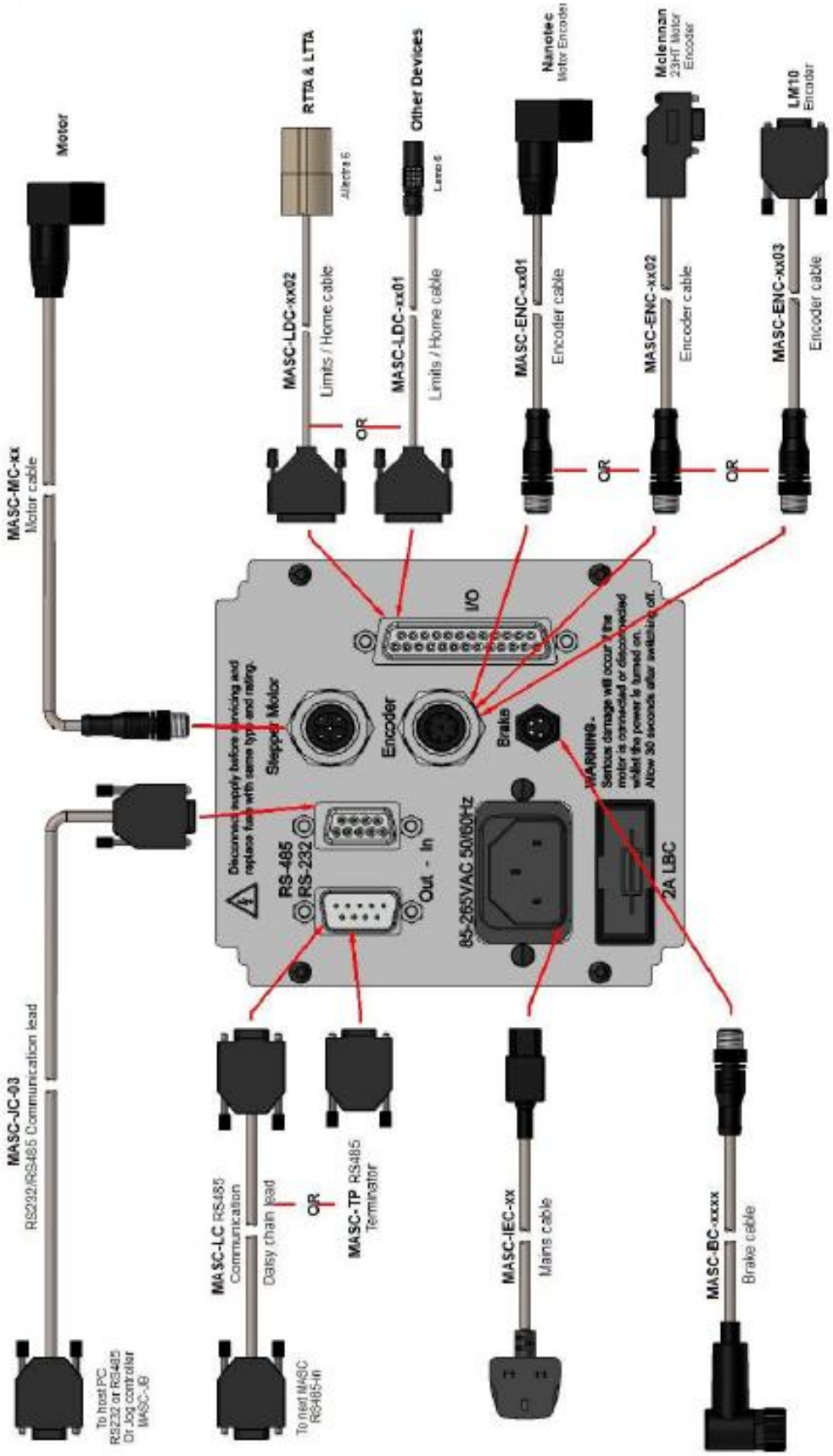
**Maximum capacity 200mA shared between this connector and the I/O connector.

RS232 / RS485 IN		
Connector type: 9 way 'D' type socket		
Terminal	**RS232 Connection**	**RS485 Connection**
1		MASC Rx+ (To host Tx+)
2	Transmit Data (To host Rx)	
3	Receive Data (To host Tx)	
4		
5	0V	
6		MASC Rx- (To host Tx-)
7		MASC Tx+ (To host Rx+)
8		MASC Tx- (To host Rx-)
9		+5V (For JOG box supply only)

RS485 OUT	
Connector type: 9 way 'D' type plug	
Terminal	**RS485 Connection**
1	Rx+
2	
3	
4	
5	
6	Rx-
7	Tx+
8	Tx-
9	

For connection to subsequent MASC units.
The last unit in the chain (including single axis solutions) must have the MASC-TP terminator plugged in to the RS-485 OUT Port.

4 Cable Configurations



5 Alarm Codes

In the event of an error, the red and green indicators on the ***MASC*** front panel will flash in alternating red-green patterns as shown below. The pattern repeats until the alarm is cleared.

Code	Error
solid green	No alarm, motor disabled
flashing green	No alarm, motor enabled
1 red 1 green	Motor stall (optional encoder only)
1 red 2 green	Move attempted while drive disabled
2 red 1 green	Ccw limit
2 red 2 green	Cw limit
3 red 1 green	Drive overheating
3 red 2 green	Internal voltage out of range
3 red 3 green	Blank Q segment
4 red 1 green	Power supply overvoltage
4 red 2 green	Power supply undervoltage
5 red 1 green	Over current / short circuit
5 red 2 green	Motor resistance out of range
6 red 1 green	Open motor winding
6 red 2 green	Bad encoder signal (optional encoder only)
7 red 1 green	Serial communication error
7 red 2 green	Flash memory error
8 red 1 green	Internal voltage out of range

6 Installation

6.1 Unpacking

The MASC unit is shipped within a specific card box with foam inserts to support the MASC. If returning the device to UHV Design for any reason, then it should be returned in its original packaging.

Do not use a knife to open the packaging as cables are also stored inside the box and may be damaged.

MASC controller should be unpacked in a suitable environment. See section 9 for environmental requirements.

6.2 Lifting and Moving

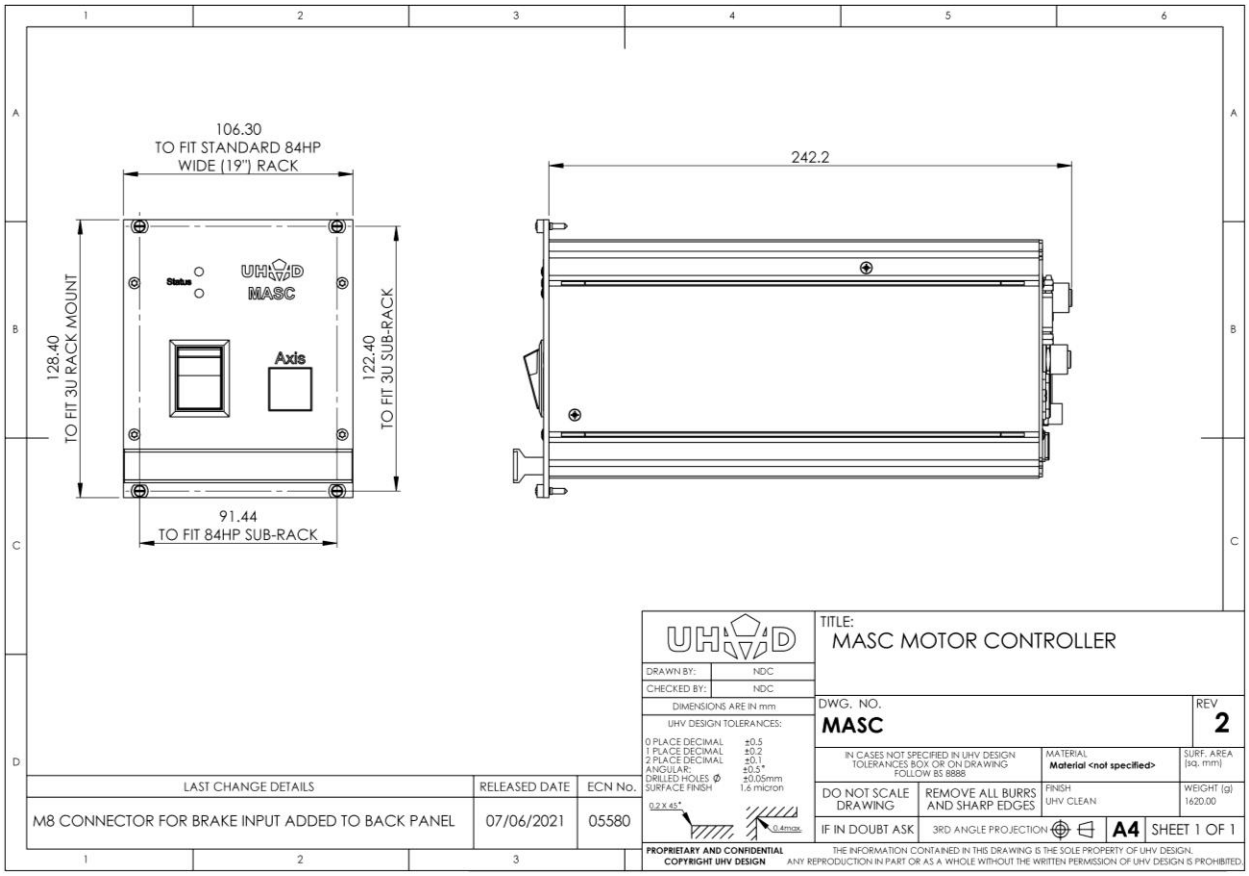
The MASC controller is weight 1.6kg. No lifting equipment is required for installation purposes.
The MASC controller should not be unnecessarily inverted or shaken to prevent damage to the unit.

6.3 Mounting

The MASC controller has been designed to be mounted in a standard 19” rack. Rack mounting blank panel kits and guide rails can be purchased separately. Contact UHV Design for details.
The MASC controller can also be mounted on a bench, provided the environmental conditions are met.

Standard mounting screws are provided as per the drawing below.

6.3.1 Drawing


106.30
TO FIT STANDARD 84HP
WIDE (19") RACK
242.2
128.40
TO FIT 3U RACK MOUNT
122.40
TO FIT 3U SUB-RACK
91.44
TO FIT 84HP SUB-RACK
Status
UHVD
MASC
Axis
LAST CHANGE DETAILS
RELEASED DATE
ECN No.
M8 CONNECTOR FOR BRAKE INPUT ADDED TO BACK PANEL
07/06/2021
05580
DRAWN BY: NDC
CHECKED BY: NDC
DIMENSIONS ARE IN mm
UHV DESIGN TOLERANCES:
0 PLACE DECIMAL ±0.5
1 PLACE DECIMAL ±0.2
2 PLACE DECIMAL ±0.1
ANGULAR: ±0.5°
DRILLED HOLES Ø ±0.05mm
SURFACE FINISH 1.6 micron
0.2 X 45°
0.4max.
TITLE:
MASC MOTOR CONTROLLER
DWG. NO.
MASC
REV
2
IN CASES NOT SPECIFIED IN UHV DESIGN TOLERANCES BOX OR ON DRAWING FOLLOW BS 8888
MATERIAL
Material <not specified>
SURF. AREA (sq. mm)
DO NOT SCALE DRAWING
REMOVE ALL BURRS AND SHARP EDGES
FINISH
UHV CLEAN
WEIGHT (g)
1620.00
IF IN DOUBT ASK
3RD ANGLE PROJECTION
A4
SHEET 1 OF 1
PROPRIETARY AND CONFIDENTIAL
COPYRIGHT UHV DESIGN
THE INFORMATION CONTAINED IN THIS DRAWING IS THE SOLE PROPERTY OF UHV DESIGN.
ANY REPRODUCTION IN PART OR AS A WHOLE WITHOUT THE WRITTEN PERMISSION OF UHV DESIGN IS PROHIBITED.

7 Links

The ***MASC*** range of packaged stepper motor drivers are based around the Applied Motion Products ST5-Q range of drives. Applied Motion Products comprehensive documentation and software applications are directly compatible with the ***MASC*** products, giving access to a powerful range of functionality and customization potential. The following references are useful places to find the technical range of literature and software offered by Applied Motion Products.

Ref #	Description and links
1	Applied Motion Products corporate website https://www.applied-motion.com/
2	ST Hardware Manual for ST5 & ST10 Q & Si drives (note that the ***MASC*** product is based on the ST5-Q range of drives with RS-485 interface and an optional Encoder interface) https://applied-motion.s3.amazonaws.com/documents/Manuals/ST5-10-QSi_Hardware%20Manual_920-0004F-opt-.pdf
3	Host Command Reference Manual for Q & SCL drives https://appliedmotion.s3.amazonaws.com/Host-Command-Reference_920-0002V.pdf
4	Applied Motion Software Downloads https://www.applied-motion.com/products/software/st-configurator https://www.applied-motion.com/products/software/q-programmer
5	Application Notes (note that the ***MASC*** product has some specific configuration constraints that need to be considered when following general Application Notes. Please request them from your supplier.) https://www.applied-motion.com/s/support/application-notes

The hyperlinks above are beyond our control and subject to change without notice, please contact UHV Design if there are any problems accessing the links.

8 Motor Selection

8.1 General Specification

Controller cannot run 3 or 5 phase stepper motors. 2 phase bipolar motors are recommended.
Controller max output current = 4A. Motor can have higher current rating but will be limited by controller output.
Controller max voltage output = 24V. Motor can have higher voltage rating but will be limited by controller output.

8.2 Integrated Sensors

The MASC controller is not designed to work with motor integrated sensors, other than shaft mounted encoders. The electrical ratings and pin-out are described in section 2.3
Typically 500 line and 1000 line encoders are specified for use.

8.3 Critical torsional Speeds

Not Applicable

8.4 Transient Torque Analysis

Not Applicable

8.5 Field Installed Components

Not Applicable

9 LIMITATIONS OF USE

This system has been designed to drive a hybrid type stepper motor. Any other use has not been considered, therefore could damage the drive and could be dangerous for the users.
Good engineering practices should be employed when using this product.
The operating temperature should be between 0ºC and 30ºC.
The system should not be exposed to vibration and mechanical shocks. This may damage the unit.
Ambient humidity should be between 30% and 70% RH.
Pollution degree 2 according to IEC 60664-1. Only non-conductive pollution, except occasional temporary conductivity caused by condensation.
Users should take suitable precautions in the application of this product, to ensure that the overall system complies with EN50081-1 and EN50082-1 (EMC directive).
These products should not be put into service until the machinery in which they are incorporated has been declared in conformity with the provisions of The Supply of Machinery (Safety) Regulations 1992 and The Supply of Machinery (Safety) (Amendment) Regulations 1994 (Machinery Directive).
Motor control systems are inherently hazardous. Even a small motor, if coupled to a leadscrew, gearbox, or any other form of mechanism that provides a mechanical advantage, can generate considerable force and could cause serious injury. Incorrect operation can also lead to damage to the motor or associated machinery.

10 DECLARATION OF CONFORMITY

EU Declaration of Conformity

Product Details

Product:	Stepper Motor Controller
Model:	MASC
Product Description:	Production sample
Specification:	230V / 50Hz

Manufacturer

Name:	UHV Design Limited
Address:	Acorn House, The Broyle, Ringmer East Sussex, England, BN8 5NN
Web site:	www.uhvdesign.com
Email:	sales@uhvdesign.com

EU Authorised Representative

Name:	Judges Scientific (Dublin) Limited
Address:	2nd Floor, 1-2 Victoria Buildings, Haddington Road, Dublin, Dublin 4. Ireland
Web site:	www.judges.uk.com
Email:	enquiries@judges.uk.com

This declaration is issued under the sole responsibility of the manufacturer.

The object of this declaration is in conformity with the essential requirements and other requirements of the relevant EU Directives (and their amendments).

Directive	Title
2014/35/EU	Low Voltage Directive (LVD)
2014/30/EU	Electromagnetic Compatibility Directive (EMC)
2011/65/EU as amended by 215/863/EU	Directive on the Restriction of the use of Certain Hazardous Substances in Electrical and Electronic Equipment (RoHS3)

Harmonised Standard		Title
LVD	BS EN IEC 61800-5-1:2023	Adjustable speed electrical power drive systems. Part 5-1: Safety Requirements – Electrical, thermal and energy
EMC	BS EN 61800-3: 2004/A1: 2012	Adjustable speed electrical power drive systems - Part 3: EMC requirements and specific test methods
	EN 61000-6-3: 2007 + A1:2011	Electromagnetic compatibility (EMC) - Part 6-3: Generic standards - Emission standard for residential, commercial and light-industrial environments
	BS EN 61800-3: 2004/A1: 2012	Adjustable speed electrical power drive systems - Part 3: EMC requirements and specific test methods
	EN 61000-6-1: 2007	Electromagnetic compatibility (EMC) - Part 6-1: Generic standards - Immunity for residential, commercial and light-industrial environments
	EN 61000-3-2: 2014	Electromagnetic compatibility (EMC). Limits. Limits for harmonic current emissions (equipment input current ≤ 16 A per phase)
	EN 61000-3-3: 2013	Electromagnetic compatibility (EMC). Limits. Limitation of voltage changes, voltage fluctuations and flicker in public low-voltage supply systems, for equipment with rated current ≤ 16 A per phase and not subject to conditional connection
RoHS3	EN 63000: 2018	Technical documentation for the assessment of electrical and electronic products with respect to the restriction of hazardous substances

Signed for and on behalf of:	UHV Design Limited	
Place of issue:	East Sussex, England	
Date of issue:	5th November 2025	j.eyres@uhvdesign.com - Eng.Director

TCFs791 Issue 1.0 Page 1 of 1

UK Declaration of Conformity

Product Details

Product:	Stepper Motor Controller
Model:	MASC
Product Description:	Production sample
Specification:	230V / 50Hz

Manufacturer

Name:	UHV Design Limited
Address:	Acorn House, The Broyle, Ringmer East Sussex, England, BN8 5NN
Web site:	www.uhvdesign.com
Email:	sales@uhvdesign.com

UK Authorised Representative

Name:	UHV Design Limited
Address:	Acorn House, The Broyle, Ringmer East Sussex, England, BN8 5NN
Web site:	www.uhvdesign.com
Email:	sales@uhvdesign.com

This declaration is issued under the sole responsibility of the manufacturer.

The object of this declaration is in conformity with the relevant UK Statutory Instruments (and their amendments).

Statutory Instrument	Title
2016 No. 1101	The Electrical Equipment (Safety) Regulations 2016 (Safety)
2016 No. 1091	The Electromagnetic Compatibility Regulations 2016 (EMC)
2012 No. 3032	The Restriction of the Use of Certain Hazardous Substances in Electrical and Electronic Equipment Regulations 2012 (RoHS)

Designated Standard		Title
Safety	BS EN IEC 61800-5-1:2023	Adjustable speed electrical power drive systems. Part 5-1: Safety Requirements – Electrical, thermal and energy
EMC	BS EN 61800-3: 2004/A1: 2012	Adjustable speed electrical power drive systems - Part 3: EMC requirements and specific test methods
	EN 61000-6-3: 2007 + A1:2011	Electromagnetic compatibility (EMC) - Part 6-3: Generic standards - Emission standard for residential, commercial and light-industrial environments
	BS EN 61800-3: 2004/A1: 2012	Adjustable speed electrical power drive systems - Part 3: EMC requirements and specific test methods
	EN 61000-6-1: 2007	Electromagnetic compatibility (EMC) - Part 6-1: Generic standards - Immunity for residential, commercial and light-industrial environments
	EN 61000-3-2: 2014	Electromagnetic compatibility (EMC). Limits. Limits for harmonic current emissions (equipment input current ≤ 16 A per phase)
	EN 61000-3-3: 2013	Electromagnetic compatibility (EMC). Limits. Limitation of voltage changes, voltage fluctuations and flicker in public low-voltage supply systems, for equipment with rated current ≤ 16 A per phase and not subject to conditional connection
RoHS	EN 63000: 2018	Technical documentation for the assessment of electrical and electronic products with respect to the restriction of hazardous substances

Signed for and on behalf of:	UHV Design Limited	
Place of issue:	East Sussex, England	
Date of issue:	5th November 2025	j.eyres@uhvdesign.com - Eng.Director

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SAFETY NOTICE

Motor drive systems are inherently hazardous. Even a small motor, if coupled to a leadscrew, gearbox, or any other form of mechanism which provides a mechanical advantage, can generate considerable force and could cause serious injury Incorrect operation can also lead to damage to the motor or associated machinery

Static sensitive devices - This unit has static sensitive devices. Observe handling precautions: Hold card by edges only. Do not touch connector pins. Ship only in anti-static packaging.

Warning - Mains voltages can be present within a system, do not adjust, service, dis-assemble or otherwise work on this system whilst power is applied.

In accordance with European directive 2002/96/EC on Waste Electrical and Electronic Equipment (WEEE), disposal of this item must be through the correct channels for WEEE. It should not be disposed of by landfill.

This item can be returned to Mclennan for end of life disposal under the WEEE directive, contact Mclennan for further information.

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